

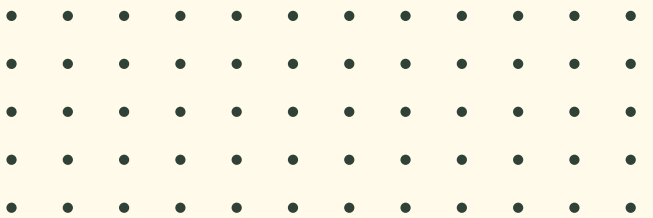
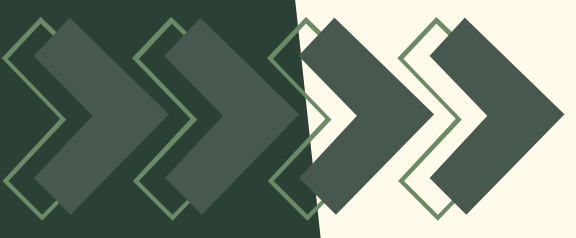
ChE : 208

Process Data Analytics & Monitoring

Multivariate Analysis on Cricket Data

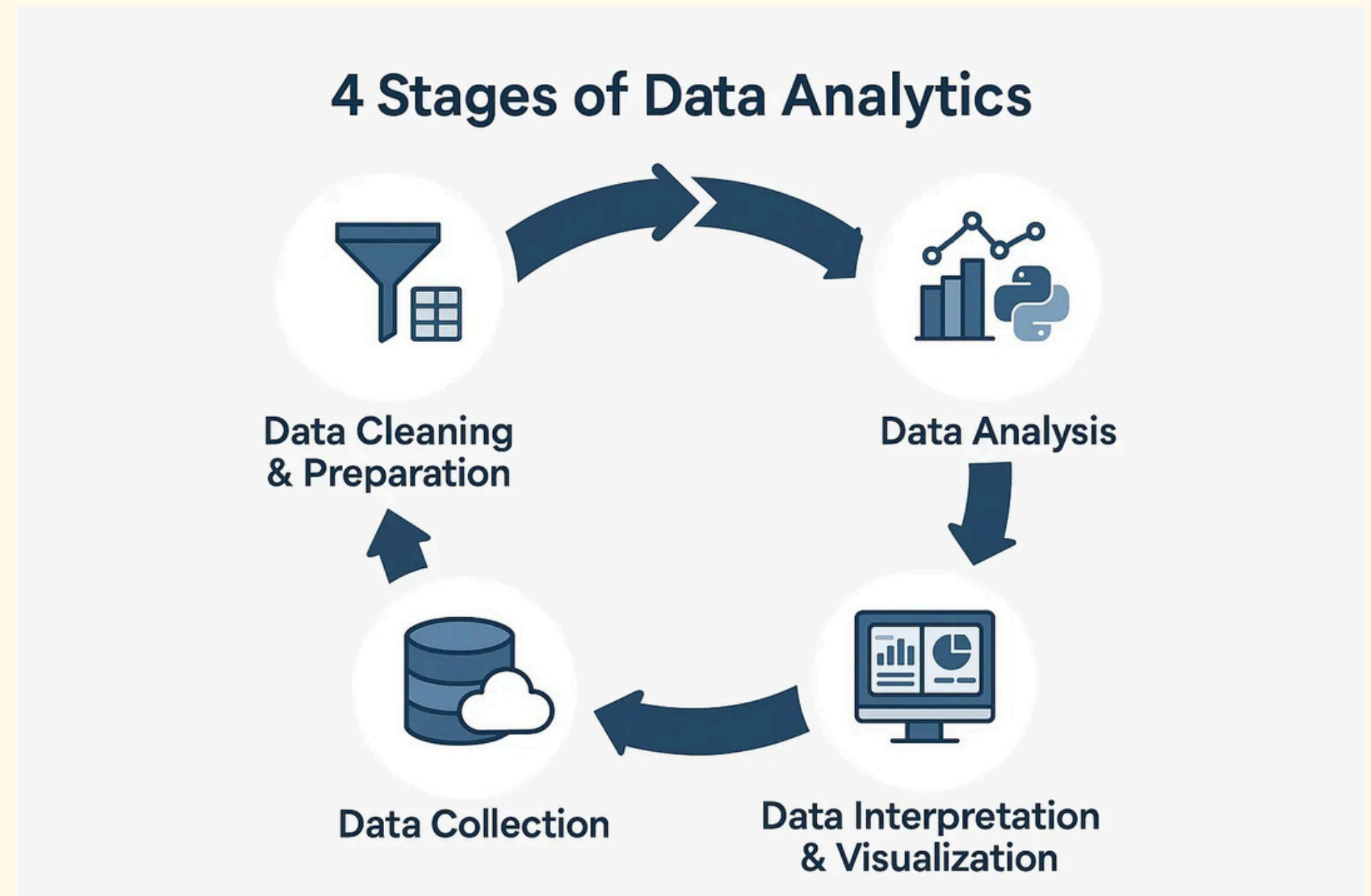


Supervisor : Dr. Abhishek Aggarwal



Member Roles

- **Vishal Shakya** : Data Pre - Processing And Time Series Analysis
- **Utkarsh Sharma** : Multivariate SPC
- **Uday Ranode** : Predictive Modelling



Overview

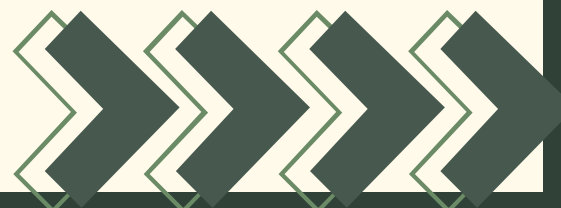
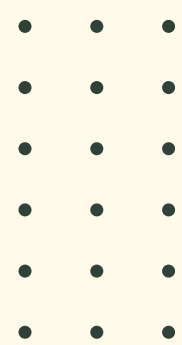


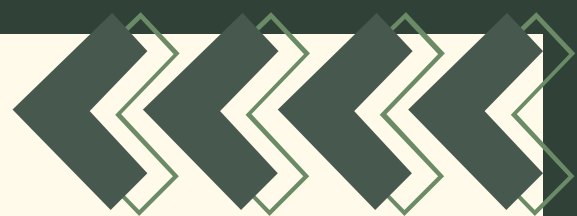
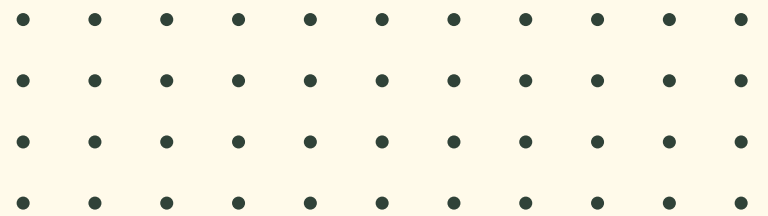
The Objective: To build a predictive model that accurately forecasts a cricket player's Points before a match begins.

The Challenge: Cricket generates massive, highly correlated datasets. With 22 players and 15+ stats each, a single match contains over 330 overlapping variables (e.g., runs, boundaries, balls faced).

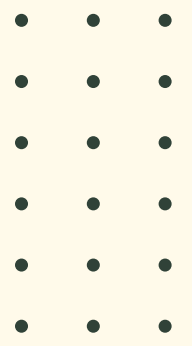
The Solution: We applied a 4-stage Process Data Analytics pipeline to clean, compress, and feed this data into an Artificial Neural Network, Ridge Regression acting as a "Soft Sensor" to predict end-of-match scores.

The Final Output: A mathematically optimized, algorithmically drafted 11-player Dream Team.





Dataset



Dataset Structure: 1 Row = 1 Complete Cricket Match. Massive Dimensionality:
Over 330 variables (columns) per single match.

Input Features (X):

Match Context: Format (ODI, T20, Test), Venue, and Match Type

Player Statistics: 14 distinct performance metrics recorded for all 22 players on the field (e.g., Runs, Wickets, 4s, 6s, Balls Played, Maidens, Catching).

Target Variable (Y): The exact points score generated by each player. This is the final output our Neural Network is trained to predict.

Data Splits: Separated into train_data (where the model learns the rules) and test_data (where the model is blindly tested).

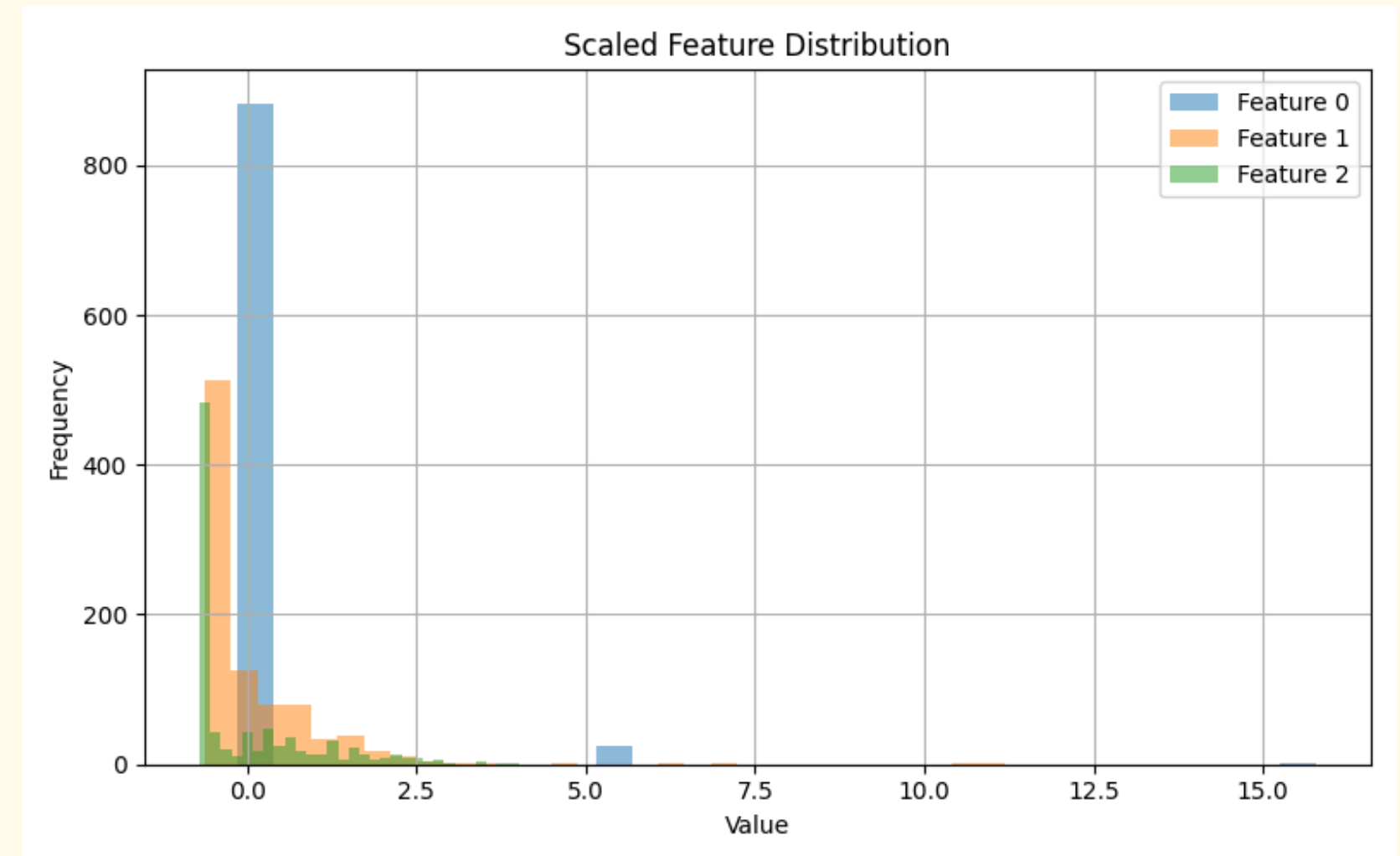
Data Preparation & Scaling

Objective: Establish a clean, uniform dataset for multivariate modeling.

Missing Data: Handled via median imputation to prevent skewed outliers from affecting baseline metrics.

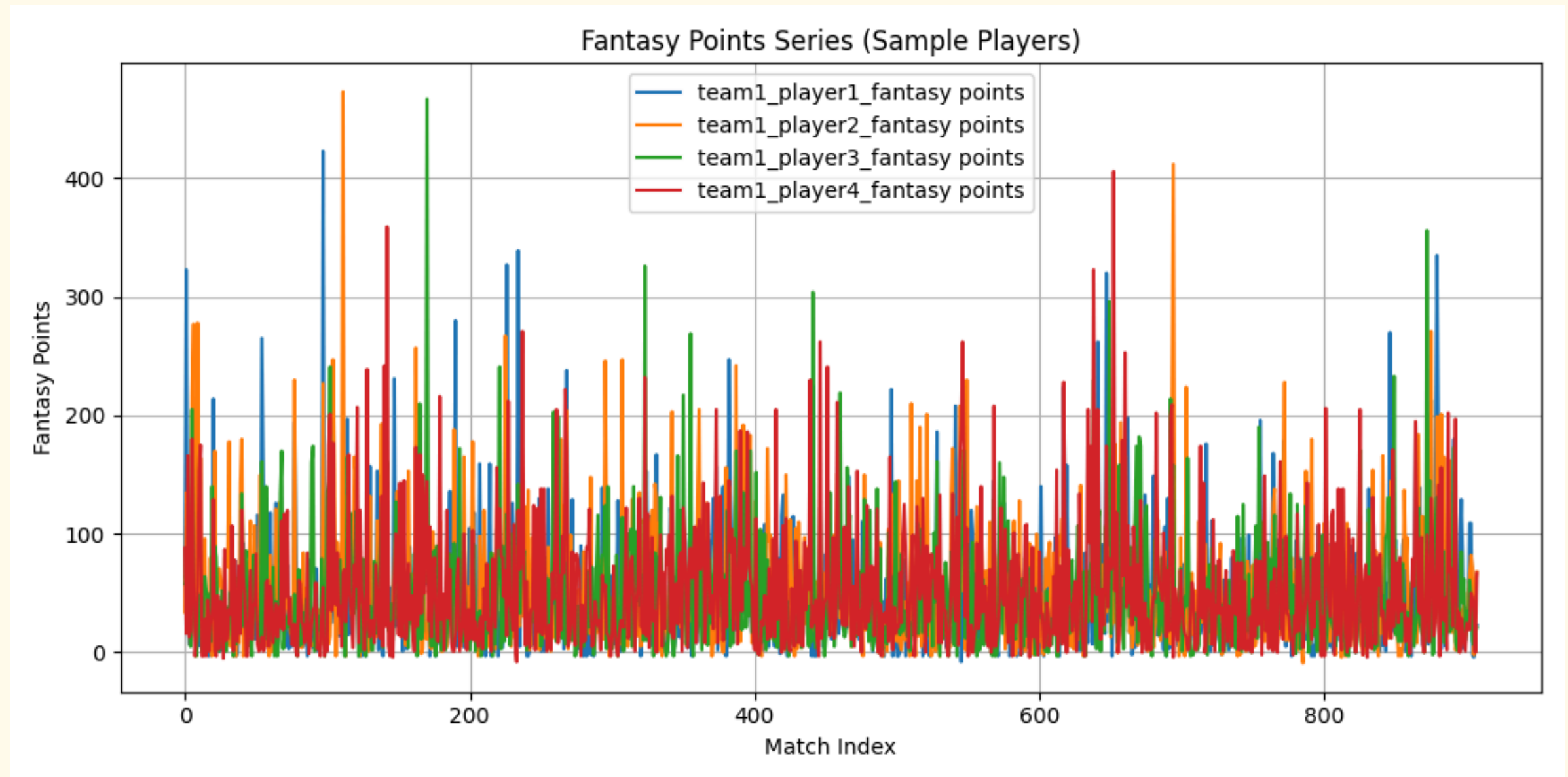
Normalization: Applied standard Z-Score scaling across all numerical features.

Variance Thresholding: Removed zero-variance features that provide no mathematical value to the models.



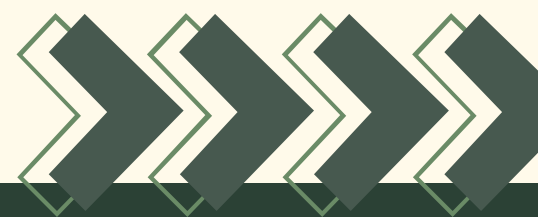
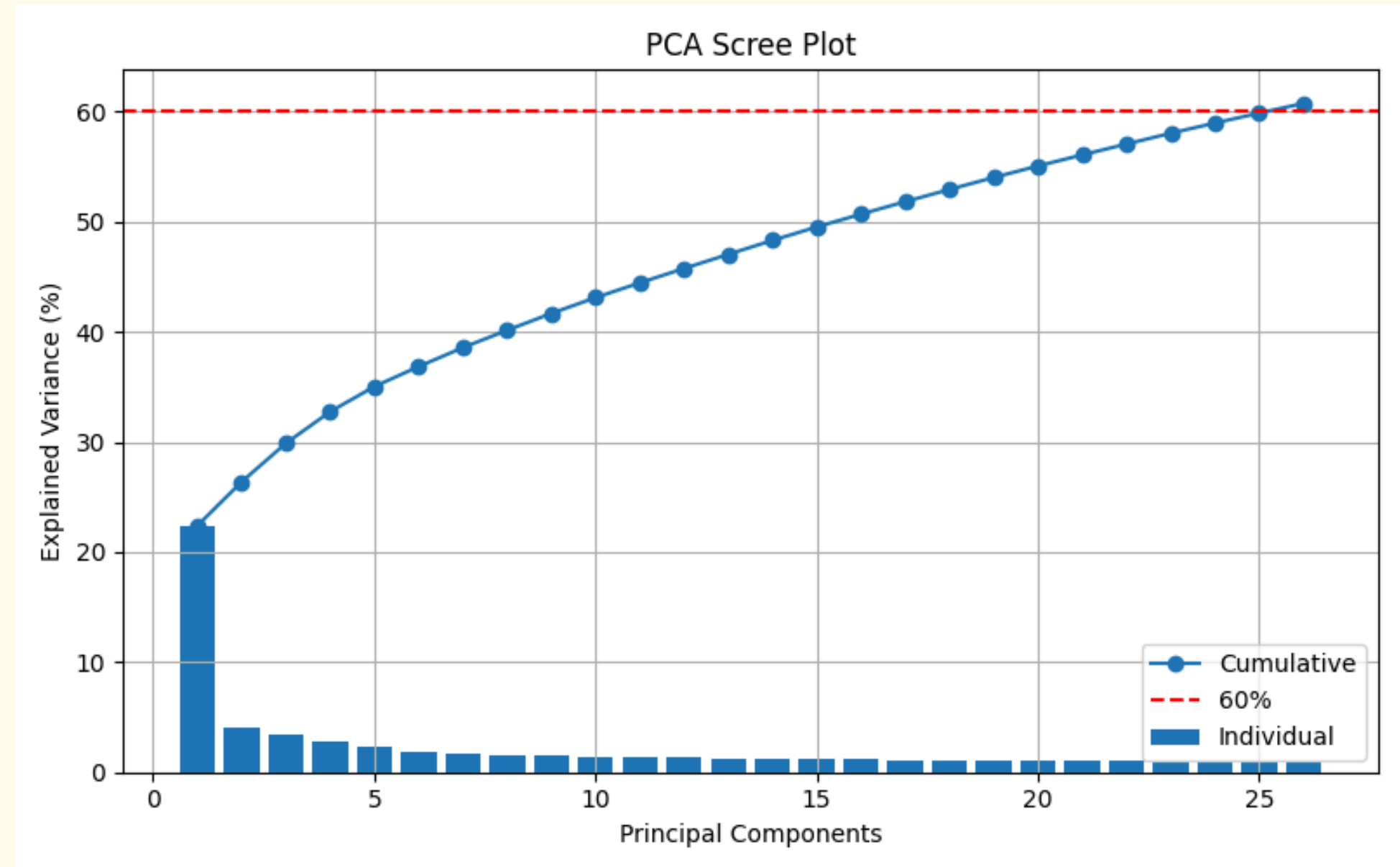
Performance Monitoring

- **Objective:** Track player form and monitor process drift over time.
- **Chronological Nature:** Match data is inherently sequential; past performance influences future predictions.
- **Validation Strategy:** Implemented a Time-Series Split for cross-validation instead of standard K-Fold to prevent data leakage.



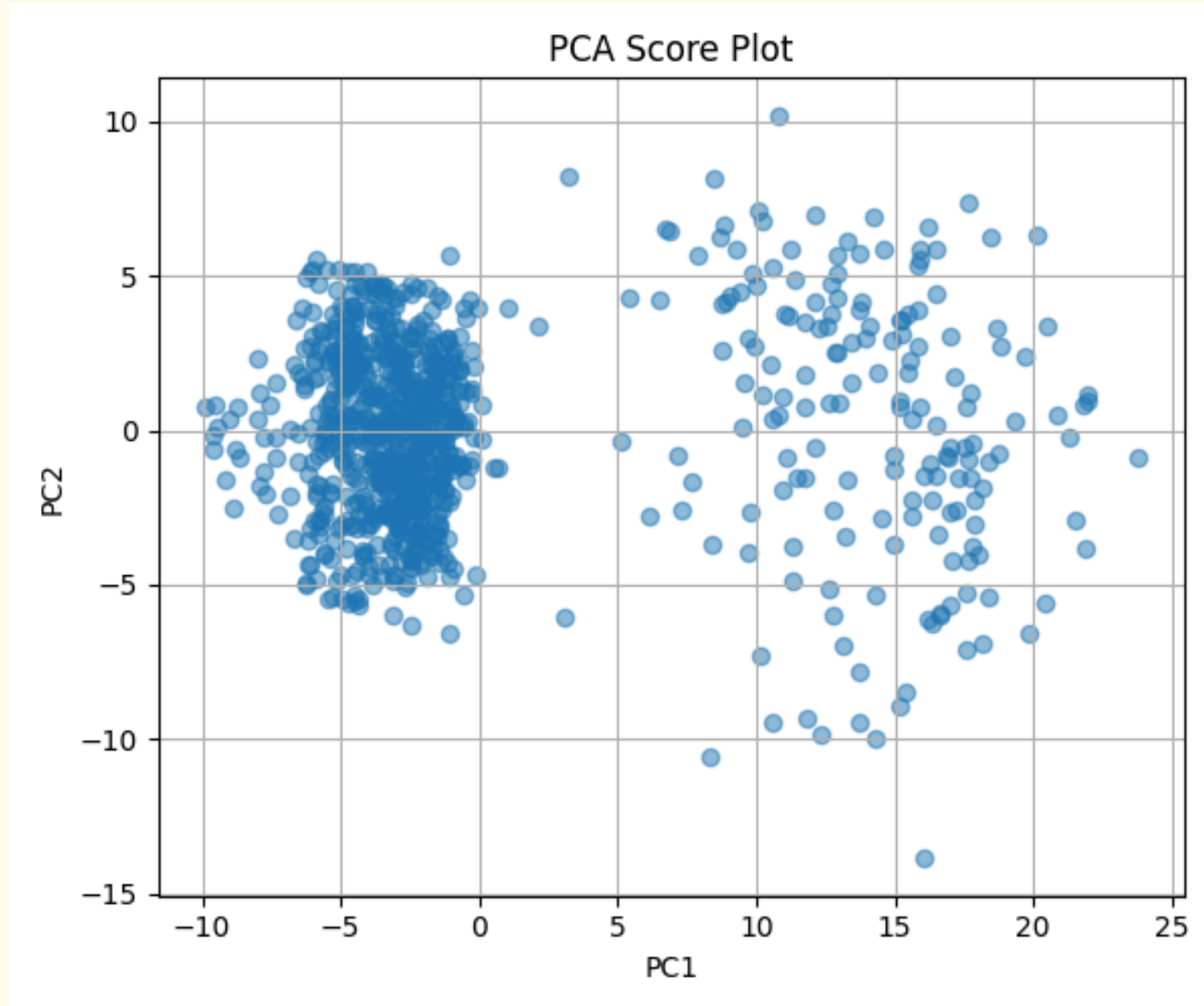
Dimensionality Reduction using PCA

- **Objective:** Simplify the massive dataset (hundreds of player stats) without losing critical variance or information.
- **PCA Scree Plot:** Identified the optimal number of Principal Components (PCs). Captured 60% of the data's variance within the first few components.



Dimensionality Reduction using PCA

- **PCA Score Plot (PC1 vs. PC2):**
Mapped the complex, multidimensional match data into a clean, reduced visual space (Score Plot).



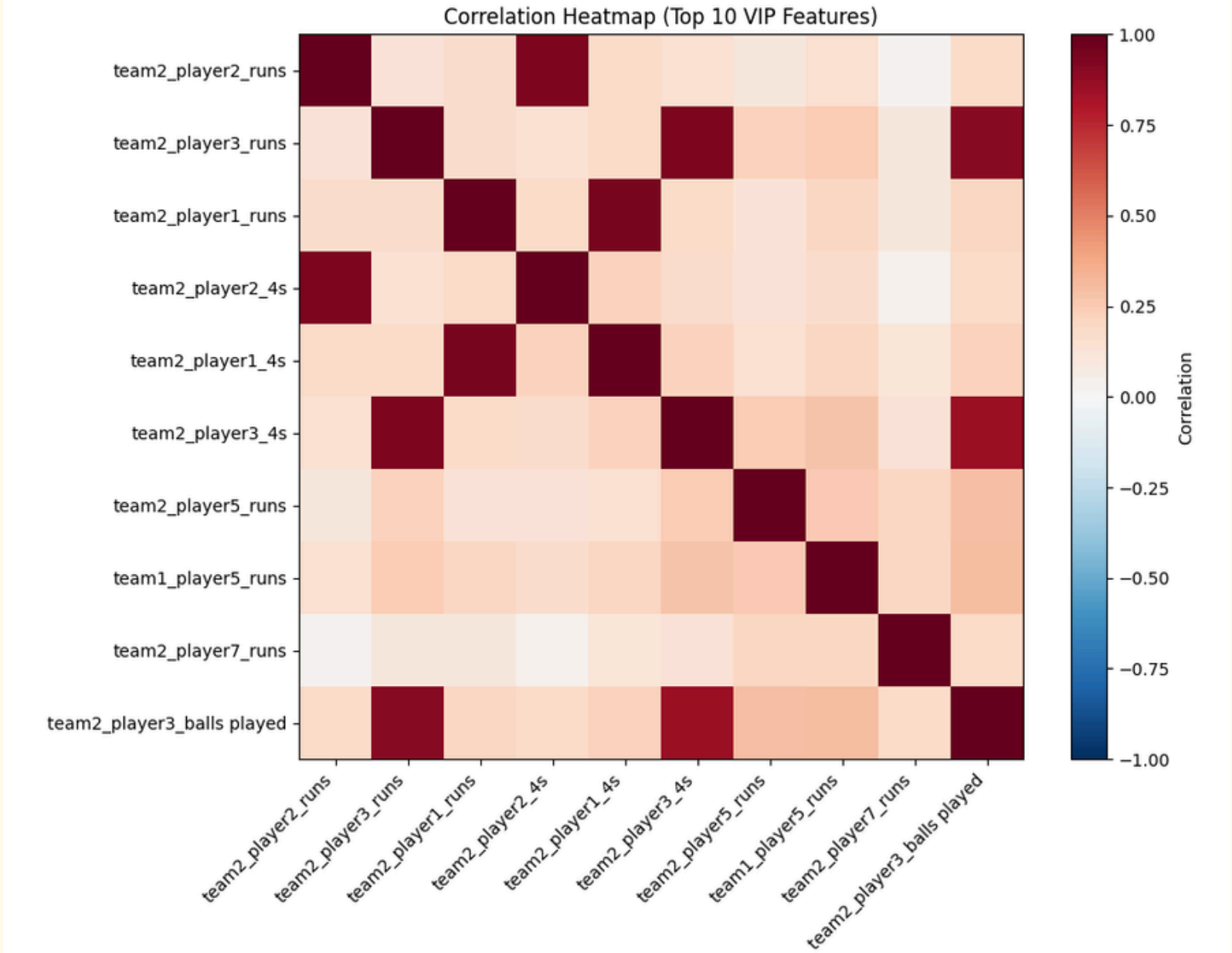
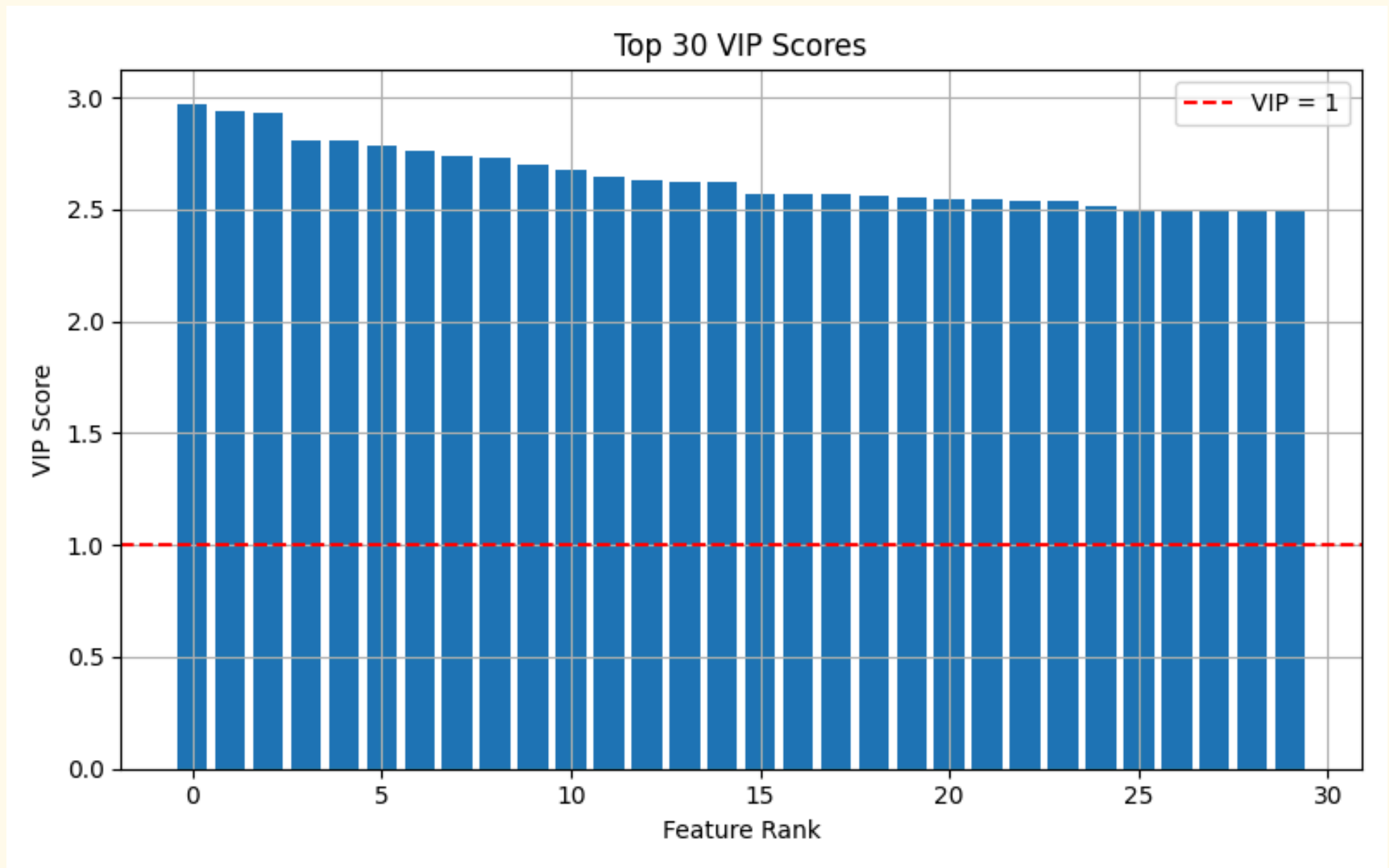
Feature Selection with PLS & VIP Scores

- **Projection to Latent Structures (PLS) model:** Mapped the relationship between player statistics (X) and our target variable, Points (Y).
- **Variable Importance in Projection (VIP):** Calculated the contribution of each individual stat to the final prediction.
- **Filtering Rule:** Applied a strict threshold of $VIP > 1.0$.
- **Result:** Filtered out statistical "noise" and isolated the "Critical-to-Quality" variables (e.g., Recent Form, Runs, Wickets) to feed into the final Machine Learning models.

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Feature Selection with PLS & VIP Scores



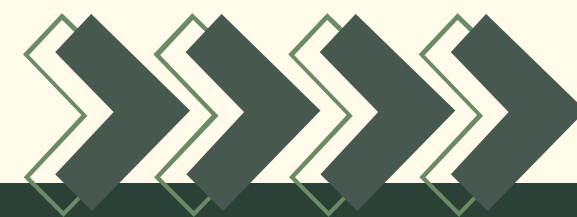
Predictive Modelling

Objective: Train and compare multiple machine learning models to predict each player's Fantasy Points using the VIP-filtered features.

4 Models were evaluated:

- K-Nearest Neighbors (KNN)
- Ridge Regression (MLR with Ridge)
- Artificial Neural Network (Soft Sensor)
- PLS Regression (Baseline)

Validation: 5-Fold Time Series Cross-Validation was used throughout to prevent data leakage and ensure real-world reliability.



Model 1: KNN

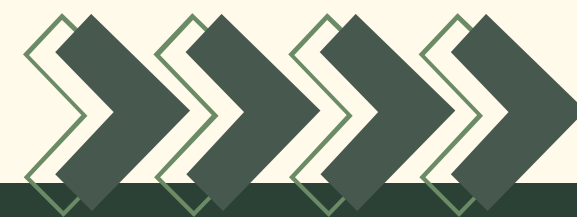
K-Nearest Neighbors (KNN)

- Input: PCA-reduced features (26 components capturing 60% variance)
- Best K selected via 5-Fold Time Series Cross-Validation

K=3 → MAE: 29.25 | K=5 → MAE: 28.12

K=7 → MAE: 27.67 | K=10 → MAE: 27.53

- Best K = 10 (lowest cross-validation MAE)
- Predicts points by averaging 10 most similar historical matches

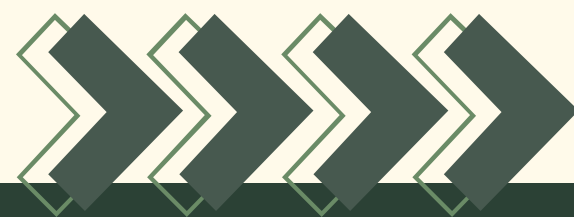


Model 2 : Ridge Regression

Ridge Regression (MLR with Ridge):

- Input: 235 VIP-selected features (out of 1153 after preprocessing)
- Ridge penalty $\alpha = 1.0$ stabilizes the model
- Prevents overfitting caused by highly correlated stats (e.g., runs, 4s, 6s are all related)
- Multi-output: predicts all 22 players simultaneously

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Model 3 : Artificial Neural Network

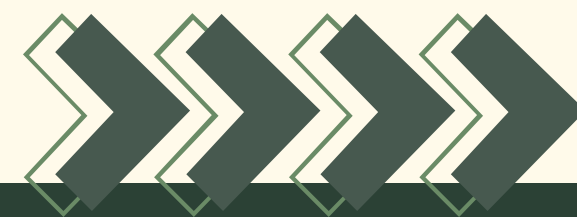
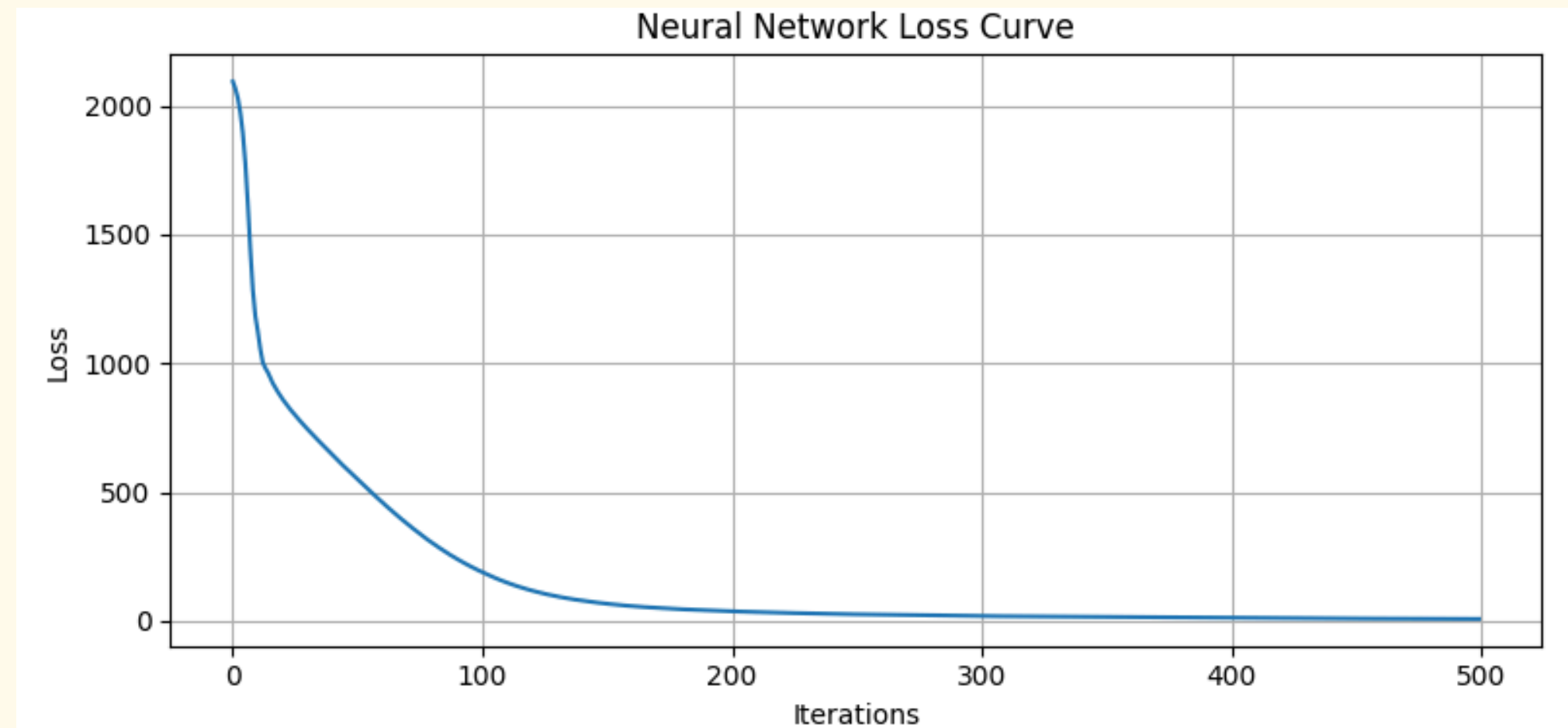
Architecture: 2 hidden layers → 128 neurons → 64 neurons → Output

Training: 500 iterations, Input = VIP-selected features

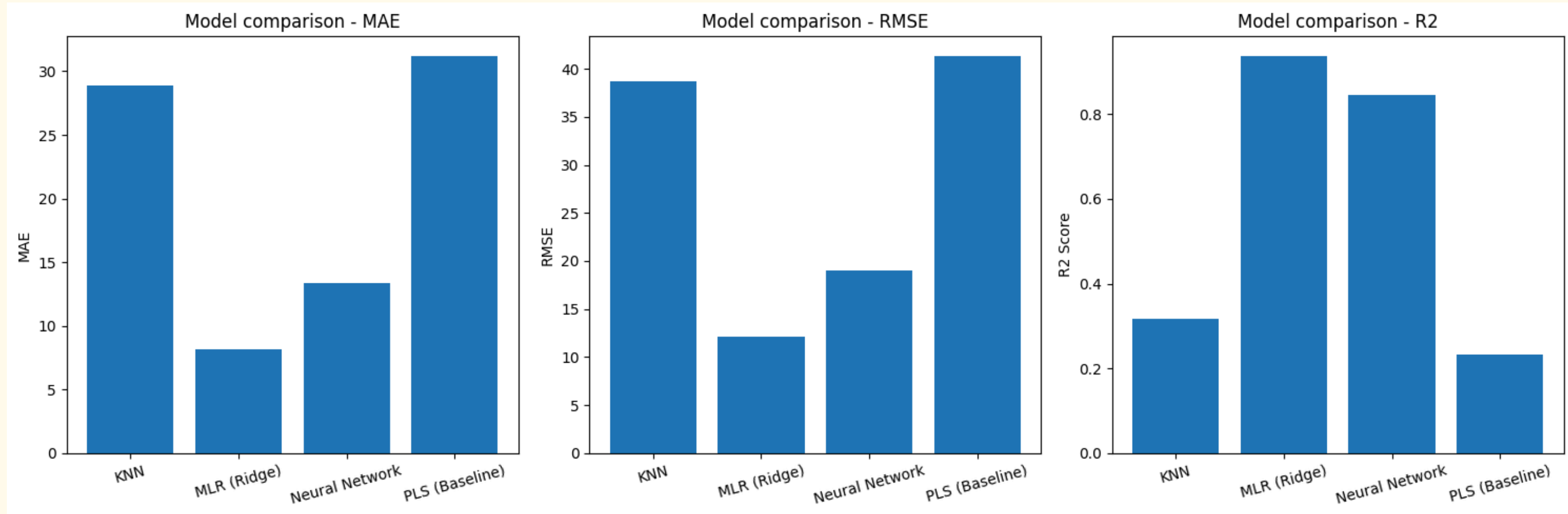
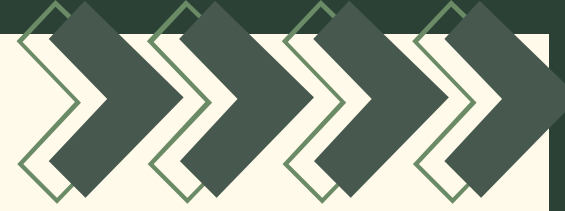
Why "**Soft Sensor**"?

- Fantasy Points are a "delayed measurement" — only known after the match ends
- The ANN estimates these points BEFORE the match using only pre-match historical stats
- This is exactly how a Soft Sensor works in process engineering — predicting a hard-to-measure output from easy-to-measure inputs

The loss curve confirms smooth convergence with no overfitting.



Model Comparison Results



Selection Criterion: Lowest MAE (Mean Absolute Error)

- Ridge Regression is the Best Model
 - $R^2 = 0.937$ means it explains 93.7% of variance in fantasy points
 - Average prediction error of only 8.16 points per player

Table 1: Model Performance Comparison

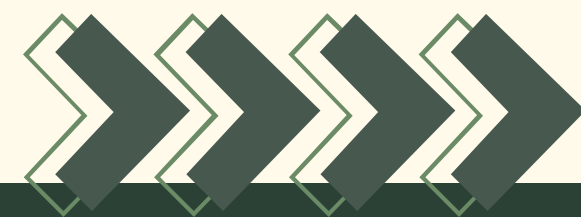
Model	MAE	RMSE	R ²
KNN	28.884182	38.746077	0.317373
Ridge	8.162378	12.097829	0.937107
Neural Net	13.355541	18.984729	0.845795
PLS	31.184599	41.320261	0.232898

Ridge model predictions, all 22 players were ranked by their Average Predicted Fantasy Points:

🔥 Best Playing XI:

	Player	Avg_Predicted_Points
0	team2_player4_fantasy points	62.618110
1	team1_player2_fantasy points	59.113322
2	team2_player3_fantasy points	56.343407
3	team1_player1_fantasy points	56.189733
4	team2_player5_fantasy points	55.272668
5	team1_player3_fantasy points	53.949150
6	team2_player1_fantasy points	53.806868
7	team1_player5_fantasy points	52.653067
8	team2_player8_fantasy points	52.324687
9	team1_player7_fantasy points	52.175488
10	team1_player4_fantasy points	51.807100

* Best Player: team2_player4_fantasy points
Predicted Avg Points: 62.61811020836356



Thank you